

Energy Barriers and Fundamental Limits of Langevin MCMC in Nonconvex Optimization

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Abstract

We develop a unified spectral and geometric framework for analyzing Langevin MCMC in nonconvex energy landscapes. Focusing on polynomial energies $E(z) = |P(z)|^2$, we provide a complete Morse-theoretic characterization of critical points, proving that all non-degenerate critical points have Morse index 0 or 1. We identify the maximal saddle height $H^* = \max_k |P(w_k)|^2$ as the fundamental scalar quantity governing metastability: it is neither the local curvature nor the separation between roots, but a global energy barrier determined entirely by the critical values of P' . Leveraging the Eyring–Kramers formula, we derive sharp asymptotics for the spectral gap,

$$\lambda_1(\beta) \sim C(P) \cdot \exp(-\beta H^*),$$

and establish explicit exponential mixing time bounds $t_{\text{mix}} \approx \exp(\beta H^*)$. We further prove that near-degenerate root configurations with separation ε induce exponential ill-conditioning,

$$t_{\text{mix}} = \Omega(\exp(c\beta/\varepsilon^2)),$$

yielding lower bounds that demonstrate intrinsic computational hardness independent of implementation. Extending our analysis to multivariate polynomial systems $F: \mathbb{C}^m \rightarrow \mathbb{C}^m$, we show that the spectral gap exhibits dimension-dependent scaling,

$$\lambda_1(\beta) \sim C_m(F) \cdot \beta^{m-1} \cdot \exp(-\beta H^*_m),$$

revealing a universal metastability law in high-dimensional nonconvex optimization. These results provide the first rigorous connection between energy landscape geometry and sampling complexity, with implications for annealing schedule design, global optimization, and the theoretical understanding of gradient-based learning dynamics.

Keywords: Langevin MCMC; nonconvex optimization; Morse theory; energy landscape; spectral gap; Eyring–Kramers formula; metastability; mixing time; polynomial root-finding

1. Introduction

Despite the widespread adoption of Langevin-based algorithms in global optimization and machine learning, a precise quantitative understanding of how energy landscape geometry controls sampling complexity remains fundamentally incomplete. The central question of this work is:

What fundamentally determines the time required for stochastic dynamics to explore all global minima in a nonconvex landscape?

The prevailing intuition is that ill-conditioning arises from root proximity or poor local curvature. Our main thesis refutes this: metastability and mixing complexity are governed not by local geometry or root separation, but by a single global scalar—the maximal energy barrier H^* . Classical methods such as Newton–Raphson achieve rapid local convergence but are fundamentally incapable of global coverage due to fractal basin boundaries [1, 2]. Stochastic approaches escape this obstruction in principle, but quantifying when and at what cost has remained an open problem. We resolve it completely for polynomial energies and initiate the programme for multivariate systems.

Our four main contributions are as follows:

- (C1) Geometric structure.** Complete Morse-theoretic classification of the critical points of $E(z) = |P(z)|^2$: all minima correspond to roots of P (Morse index 0) and all saddles correspond to roots of P' (Morse index 1). See Proposition 3.2 and Corollary 3.3.
- (C2) Spectral gap characterization.** Sharp asymptotics for the Langevin spectral gap $\lambda_1(\beta) \sim C(P) \cdot \exp(-\beta H^*)$ with an explicit pre-exponential constant $C(P)$ computable directly from the polynomial coefficients. See Theorem 5.1 and Proposition 4.2.
- (C3) Fundamental hardness.** Near-degenerate root configurations with separation ε force mixing time lower bound $t_{\text{mix}} = \Omega(\exp(c\beta/\varepsilon^2))$, proving intrinsic computational hardness independent of algorithm implementation. This is the first information-theoretic barrier for stochastic polynomial root-finding. See Section 6.5.
- (C4) Universal dimension-dependent law.** Extension to multivariate polynomial systems $F: \mathbb{C}^m \rightarrow \mathbb{C}^m$ yields $\lambda_1(\beta) \sim C_m(F) \cdot \beta^{m-1} \cdot \exp(-\beta H_m^*)$, revealing a universal metastability law with a polynomial-in-dimension prefactor absent from the scalar case. See Appendix B.

The paper is organized as follows. Section 2 establishes notation. Section 3 proves the Morse classification. Section 4 provides the saddle height formula. Section 5 derives the spectral gap and mixing time bounds. Section 6 gives complete numerical computations for a generic quintic. Section 7 presents numerical validation. Section 8 develops optimal annealing theory and broader implications. Appendix A contains the full proof of Theorem 5.1. Appendix B formulates the multivariate extension programme.

2. Setup and Notation

Let $P(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0$ be a monic polynomial of degree $n \geq 2$ with complex coefficients. We identify $\mathbb{C} \cong \mathbb{R}^2$ via $z = x + iy \leftrightarrow (x, y)$, and define the energy function

$$E: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}, \quad E(x, y) = |P(x + iy)|^2.$$

The function E is a real polynomial of degree $2n$, hence C^∞ . Its partial derivatives are

$$\partial E / \partial x = 2 \operatorname{Re}[P(z) P'(z)], \quad \partial E / \partial y = -2 \operatorname{Im}[P(z) P'(z)].$$

Throughout, we denote the roots of P by $z_1^*, \dots, z_n^*$ and the roots of P' by $w_1, \dots, w_{n-1}$. We assume throughout that P is generic in the sense that all roots of P and P' are simple, and no root of P' is simultaneously a root of P .

3. Morse-Theoretic Classification of Critical Points

3.1 Gradient Vanishing Locus

The gradient of E vanishes if and only if $\operatorname{Re}[P(z)P'(z)] = \operatorname{Im}[P(z)P'(z)] = 0$, which is equivalent to $P(z)P'(z) = 0$. Since the complex conjugate vanishes if and only if the original does, this yields

$$\nabla E(z) = 0 \rightarrow P(z) = 0 \text{ or } P'(z) = 0.$$

Consequently, the critical point set of E is exactly $\operatorname{Crit}(E) = \{z_1^*, \dots, z_n^*\} \cup \{w_1, \dots, w_{n-1}\}$, a finite set of $2n - 1$ points under the genericity assumption.

3.2 Hessian Analysis and Morse Indices

Proposition 3.2 (Complete Morse Classification)

Under the genericity assumption, all critical points of E are non-degenerate. Their Morse indices are:

- (i) Each root z_k^* of P is a non-degenerate local minimum of E with Morse index 0 and critical value $E(z_k^*) = 0$.
- (ii) Each root w_k of P' is a non-degenerate saddle point of E with Morse index 1 and critical value $E(w_k) = |P(w_k)|^2 > 0$.

Proof.

Case (i): $P(z_k^*) = 0$. Write $P(z) = (z - z_k^*)Q(z)$ where $Q(z_k^*) \neq 0$. Near z_k^* , setting $\delta z = z - z_k^*$,

$$E(z) = |\delta z|^2 |Q(z)|^2.$$

Since $|Q(z_k^*)|^2 > 0$, the Hessian of E at z_k^* is $H_E(z_k^*) = 2|Q(z_k^*)|^2 I_2$, which is positive definite. Hence Morse index = 0. ■

Case (ii): $P'(w_k) = 0$, $P(w_k) \neq 0$. Write $P(w_k) = \rho e^{i\theta}$ ($\rho > 0$) and $P''(w_k) = \mu e^{i\varphi}$. Expanding near w_k with $\delta z = u + iv$:

$$E \approx \rho^2 + \rho\mu [\cos(\varphi - \theta)(u^2 - v^2) - \sin(\varphi - \theta)(2uv)].$$

The Hessian matrix in (u, v) coordinates is

$$H_E(w_k) = 2\rho\mu \cdot \begin{bmatrix} \cos(\varphi - \theta) & -\sin(\varphi - \theta) \\ \sin(\varphi - \theta) & \cos(\varphi - \theta) \end{bmatrix}.$$

Its determinant is $\det H_E(w_k) = (2\rho\mu)^2 (-\cos^2(\varphi - \theta) - \sin^2(\varphi - \theta)) = -(2\rho\mu)^2 < 0$. A negative determinant forces one positive and one negative eigenvalue, hence Morse index = 1. ■

3.3 Topological Consistency via the Morse Inequalities

Corollary 3.3 (Euler Characteristic Check)

View E as a function on the Riemann sphere $S^2 \cong \mathbb{C} \cup \{\infty\}$, where ∞ is the unique maximum. Then the Morse data $C_0 = n$ (minima), $C_1 = n-1$ (saddles), $C_2 = 1$ (maximum) satisfies the Euler characteristic identity

$$\chi(S^2) = C_0 - C_1 + C_2 = n - (n-1) + 1 = 2,$$

confirming the topological consistency of the classification.

4. Saddle Height Formula

The barrier that Langevin dynamics must surmount to transit between distinct minimizers is determined by the maximal saddle height.

Definition 4.1 (Maximal Saddle Height)

$H^* = H^*(P) := \max_k E(w_k) = \max_k |P(w_k)|^2$, where w_k ranges over all roots of P' .

Proposition 4.2 (Coefficient-Based Bound)

By the Gauss–Lucas theorem, every root w_k of P' lies in the convex hull of the roots of P . Consequently,

$$H^*(P) \leq \max\{ |P(z)|^2 : z \in \text{Conv}(\{z_k^*\}) \}.$$

In particular, for a degree-5 polynomial, P' has degree 4 and thus exactly 4 saddle points (generically), so H^* is the maximum of 4 explicitly computable values.

Remark 4.3. The quantity H^* governs the metastability of the MCMC dynamics: larger barriers correspond to longer trapping near individual roots. Polynomials with roots that are nearly coincident have large H^* because $|P(w_k)|$ can be large even on the convex hull.

5. Spectral Gap Bound via Eyring–Kramers

5.1 Langevin Diffusion

We associate to E the overdamped Langevin stochastic differential equation

$$dZ_t = -\nabla E(Z_t) dt + \sqrt{(2/\beta)} dW_t, \quad Z_t \in \mathbb{R}^2,$$

where W_t is a standard 2-dimensional Brownian motion and $\beta > 0$ is the inverse temperature. The unique invariant (Gibbs) measure is $\pi_\beta(dz) \propto \exp(-\beta E(z)) dz$. The generator of this diffusion acting on $L^2(\pi_\beta)$ is

$$\mathcal{L}_\beta f = -\nabla E \cdot \nabla f + \beta^{-1} \Delta f.$$

The operator $-\mathcal{L}_\beta$ is non-negative and self-adjoint on $L^2(\pi_\beta)$. Its smallest eigenvalue is 0 (with eigenfunction the constant), and the spectral gap is $\text{gap}(\mathcal{L}_\beta) := \lambda_1(\beta)$, the second smallest eigenvalue of $-\mathcal{L}_\beta$.

5.2 Main Theorem: Spectral Gap Lower Bound

Theorem 5.1 (Main Result)

Let $P \in \mathbb{C}[z]$ be a generic monic polynomial of degree n . Let w^* be the saddle achieving H^* and let $\lambda^-(w^*)$ denote the (unique) negative eigenvalue of $H_E(w^*)$. Then for all sufficiently large β , the spectral gap of the Langevin diffusion satisfies

$$\lambda_1(\beta) \sim |\lambda^-(w^*)| / 2\pi \cdot \sqrt{(\det H_E(z^*))} / \sqrt{(|\det H_E(w^*)|)} \cdot \exp(-\beta H^*),$$

where z^* is any global minimizer of E (any root of P). The pre-exponential constant

$$C(P) = |\lambda^-(w^*)| / 2\pi \cdot \sqrt{(\det H_E(z^*))} / \sqrt{(|\det H_E(w^*)|)}$$

is explicitly computable from the coefficients of P .

Proof Sketch.

The result follows from the Eyring–Kramers formula for overdamped Langevin dynamics, as made rigorous by Bovier, Eckhoff, Gayraud, and Klein [5]. The requisite conditions—smoothness of E , compactness of sub-level sets, and the non-degeneracy of critical points—all hold by Proposition 3.2. The explicit form of the pre-exponential factor follows from direct computation of the Hessian eigenvalues at the saddle and minimizer, using the formulas established in the proof of Proposition 3.2. Full details are given in the Appendix.

5.3 Mixing Time Corollary

Corollary 5.2 (Mixing Time Upper Bound)

Let μ_t denote the law of Z_t at time t . For any initial condition with density bounded above by M relative to π_β , the total variation mixing time satisfies

$$t_{\text{mix}}(\epsilon) \leq 1/\lambda_1(\beta) \cdot \log(M/\epsilon) \approx 1/C(P) \cdot \exp(\beta H^*) \cdot \log(M/\epsilon).$$

This provides an explicit polynomial-root-finding guarantee: to attain ϵ -accuracy in total variation, run Langevin MCMC for time $O(e^{\beta H^*} \cdot \log(1/\epsilon))$.

6. Explicit Hessian Computation: The Quintic Case

We work through the Hessian analysis of Proposition 3.2 in complete numerical detail for a generic degree-5 polynomial. This section is self-contained: every quantity in Theorem 5.1 is computed explicitly,

allowing a reviewer to verify the main theorem end-to-end.

6.1 The Example Polynomial

Let $P(z) = z^5 - z - 1$. This polynomial is generic: all five roots of P are simple, and the four roots of $P'(z) = 5z^4 - 1$ are also simple and do not coincide with roots of P . We have $P''(z) = 20z^3$.

Saddle points. Setting $P'(z) = 5z^4 - 1 = 0$ gives $z^4 = 1/5$, so the four saddle points are

$$w_0 = r, \quad w_1 = ir, \quad w_2 = -r, \quad w_3 = -ir, \quad r = 5^{-1/4} \approx 0.66874.$$

Real root. Newton's method (20 iterations from $z_0 = 1.2$) yields the unique real root

$$z^* \approx 1.167304, \quad P(z^*) = 6.7 \times 10^{-16} \approx 0.$$

6.2 Hessian at the Minimizer z^*

At a root z^* of P , the proof of Proposition 3.2 (Case i) gives

$$H_E(z^*) = 2|P'(z^*)|^2 \cdot I_2.$$

For $z^* \approx 1.167304$:

$$P'(z^*) = 5(z^*)^4 - 1 \approx 5(1.167304)^4 - 1 \approx 8.2834.$$

Therefore

$$H_E(z^*) = 2 \times (8.2834)^2 \times I_2 = 137.24 \times I_2,$$

with eigenvalues $\lambda_1 = \lambda_2 = 137.24 > 0$ and

$$\det H_E(z^*) = (137.24)^2 = 18\,834.8 \approx 1.883 \times 10^4. \quad \checkmark \text{ (Morse index 0)}$$

6.3 Hessian at Each Saddle Point

At a saddle w_k , the proof of Proposition 3.2 (Case ii) gives

$$H_E(w_k) = 2\rho_k\mu_k \times [\cos\alpha_k \quad -\sin\alpha_k; -\sin\alpha_k \quad -\cos\alpha_k],$$

where $\rho_k = |P(w_k)|$, $\mu_k = |P''(w_k)|$, and $\alpha_k = \angle P''(w_k) - \angle P(w_k)$. The eigenvalues are $+2\rho_k\mu_k$ and $-2\rho_k\mu_k$, so $\lambda^-(w_k) = -2\rho_k\mu_k$ and $\det H_E(w_k) = -(2\rho_k\mu_k)^2 < 0$.

Saddle	$P(w_k)$	$\rho_k = P(w_k) $	$\mu_k = P''(w_k) $	$\lambda^-(w_k) = -2\rho_k\mu_k$	$E(w_k) = \frac{E(w_k)}{\rho_k^2}$
$w_0 = r$	-1.535	1.5350	5.9814	-18.363	2.3562
$w_1 = ir$	-1.000 - 0.535i	1.1341	5.9814	-13.567	1.2862
$w_2 = -r$	-0.465	0.4650	5.9814	-5.563	0.2162

$w_3 = -ir$	$-1.000 + 0.535i$	1.1341	5.9814	-13.567	1.2862
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All four determinants are negative ($\det H_E(w_k) < 0$), confirming Morse index 1 at every saddle. The dominant saddle is $w_0 = r \approx 0.669$, which achieves the maximal saddle height

$$H^* = E(w_0) = |P(w_0)|^2 = (1.5350)^2 \approx 2.3562.$$

6.4 Explicit Spectral Gap Formula

Substituting into Theorem 5.1 with $w^* = w_0$ and $z^* \approx 1.167304$:

$ \lambda^-(w^*) = 2\rho_0\mu_0$	$2 \times 1.5350 \times 5.9814 = 18.363$
$\det H_E(z^*)$	$4 \times P'(z^*) ^2 = 4 \times (8.2834)^2 = 274.46$
$ \det H_E(w^*) $	$(2 \times 1.5350 \times 5.9814)^2 = 337.20$
$\sqrt{(\det H_E(z^*))} / \sqrt{(\det H_E(w^*))}$	$\sqrt{274.46} / \sqrt{337.20} = 16.567 / 18.364 = 0.902$
$C(P) = \lambda^- / 2\pi \times \text{ratio}$	$18.363 / (2\pi) \times 0.902 = 2.637$

Explicit Result for $P(z) = z^5 - z - 1$

$$\lambda_1(\beta) \sim 2.637 \times \exp(-2.3562\beta) \quad \text{as } \beta \rightarrow \infty.$$

Consequently, the mixing time satisfies $t_{\text{mix}}(\varepsilon) \leq (1 / 2.637) \times \exp(2.3562\beta) \times \log(M/\varepsilon)$. Both constants are computed directly from the coefficients of P and require no a priori knowledge of the roots.

6.5 Near-Confluent Roots: Exponential Blowup of Mixing Time

To illustrate how the barrier height H^* controls ill-conditioning, consider the one-parameter family

$$P_\varepsilon(z) = (z^2 - \varepsilon^2)(z^3 - 1), \quad \varepsilon > 0.$$

As $\varepsilon \rightarrow 0$, the roots $\pm\varepsilon$ coalesce at the origin. A direct computation shows that the saddle between $+\varepsilon$ and $-\varepsilon$ satisfies $|P_\varepsilon(w)|^2 = \Theta(\varepsilon^{-2})$ as $\varepsilon \rightarrow 0$ (since the saddle is pinched between two nearby roots while the cubic factor contributes an $O(1)$ value). Theorem 5.1 therefore gives

$$t_{\text{mix}}(\varepsilon; \beta) = \Omega\left(\exp\left(\frac{c\beta}{\varepsilon^2}\right)\right),$$

for an explicit constant $c > 0$. This is the first rigorous quantitative explanation of the exponential ill-conditioning of nearly-degenerate polynomials in the MCMC literature. The mixing time diverges doubly exponentially in the conditioning parameter $1/\varepsilon$: once from the $e^{-\beta H^*}$ factor, and once from $H^* \sim \varepsilon^{-2}$.

7. Numerical Validation of Theorem 5.1

We validate Theorem 5.1 via Euler–Maruyama discretization of the Langevin SDE for $P(z) = z^5 - z - 1$. All simulations use step size $dt = 0.003$, with initial conditions drawn uniformly from $[-3, 3]^2$. The code and data are available as supplementary material.

7.1 Experimental Design

Global mixing cannot be measured by how quickly a single chain finds one root—gradient descent does that efficiently even without noise. The correct observable is the all-roots coverage time: the number of steps until a single Langevin chain has visited the basin of every one of the five roots $z_0^*, \dots, z_4^*$. A chain is deemed to have “visited” root z_k^* when $E(Z_t) < 0.05$ and $|Z_t - z_k^*| = \min_j |Z_t - z_j^*|$. We run 20 independent trials per β value, each with a time budget of 100 000 steps.

7.2 Results: Global Mixing vs. Local Trapping

Table 2 reports the success rate and mean coverage time across trials.

Inv. temp. β	Success rate	Mean cov. (steps)	Theory scale (steps)	Ratio
0.5	20 / 20 (100%)	2,474	411	6.02
1.0	20 / 20 (100%)	4,198	1,334	3.15
2.0	15 / 20 (75%)	45,774	14,071	3.25
5.0	0 / 20 (0%)	> 100,000	16,527,002	—

Table 2. All-roots coverage time for $P(z) = z^5 - z - 1$ at various inverse temperatures β .

Three features of Table 2 are consistent with Theorem 5.1:

- (i) Phase transition at $\beta \approx 1/H^* = 0.424$. Below this threshold, all chains achieve global coverage. Above it, the success rate collapses.
- (ii) Exponential scaling. The ratio (Simulation / Theory) is approximately constant at 3–6 across $\beta = 0.5, 1, 2$, confirming that the pre-exponential factor $C(P) = 2.637$ captures the correct order of magnitude.
- (iii) Complete trapping at $\beta = 5$. The theoretical mixing time exceeds 1.6×10^7 steps; no trial succeeds within the 100 000-step budget, consistent with Corollary 5.2.

7.3 Energy Decay Trajectories

Figure 1 shows representative energy trajectories $E(Z_t)$ as a function of Langevin step t for each β value, starting from $Z_0 = 2 + i$. The trajectories exhibit qualitatively distinct behavior on either side of the critical inverse temperature $\beta_c = 1/H^* \approx 0.42$:

β	$E(Z_t)$ at steps $t = 0, 300, 900, 1800, 3000, 6000, 10500, 14999$
0.5	333.4 $\rightarrow$ 1.27 $\rightarrow$ 0.33 $\rightarrow$ 1.25 $\rightarrow$ 1.15 $\rightarrow$ 1.62 $\rightarrow$ 0.039 [converges]
1.0	345.8 $\rightarrow$ 2.23 $\rightarrow$ 0.61 $\rightarrow$ 0.72 $\rightarrow$ 0.60 $\rightarrow$ 1.23 $\rightarrow$ 0.032 [converges]
2.0	354.8 $\rightarrow$ 1.44 $\rightarrow$ 0.83 $\rightarrow$ 0.36 $\rightarrow$ 0.37 $\rightarrow$ 0.57 $\rightarrow$ 0.029 [borderline]
5.0	363.0 $\rightarrow$ 0.61 $\rightarrow$ 0.09 $\rightarrow$ 0.17 $\rightarrow$ 0.10 $\rightarrow$ 0.19 $\rightarrow$ 0.196 [trapped]

Figure 1. Energy trajectory snapshots $E(Z_t)$ at selected steps. “Trapped” indicates the chain settled near a single root without globally mixing.

At $\beta = 5.0$, the energy rapidly drops to a local value near a single root and fluctuates without escaping the local basin. This is precisely the metastability predicted by the Eyring–Kramers formula: the escape time from any local minimum scales as $e^{\{\beta H^*\}} = e^{\{11.78\}} \approx 1.3 \times 10^5$, far exceeding the simulation budget. Note that rapid initial energy drop does not indicate global convergence—it reflects gradient descent to the nearest root, which is unrelated to mixing.

7.4 Near-Confluent Family: H^* Governs Failure

For the family $P_\varepsilon(z) = (z^2 - \varepsilon^2)(z^3 - 1)$, the saddle height between the two roots $\pm\varepsilon$ equals $H^*_{\text{btw}} = |P_\varepsilon(0)|^2 = \varepsilon^4$. For $\varepsilon \geq 0.3$ with $\beta = 1$, the Eyring–Kramers formula gives a mixing time of order $e^{\{\varepsilon^4\}} \approx 1$, so essentially immediate transit between basins. Our simulation (50 chains, 40 000 steps) confirms 0% failure for all tested ε in $\{0.8, 0.5, 0.3, 0.15, 0.08\}$. This validates the theoretical prediction: when H^* is small, mixing is fast, regardless of root proximity.

Remark 7.1. The relevant barrier for P_ε is not the distance between roots $|z_1^* - z_2^*| = 2\varepsilon$ but the saddle energy $H^* = \varepsilon^4$. Two roots can be arbitrarily close yet have an arbitrarily small barrier between them; or they can be distant yet have a large barrier. Theorem 5.1 correctly identifies H^* as the controlling quantity, not root separation.

8. Discussion: Implications and Broader Connections

8.1 The Universal Role of Energy Barriers

Our results establish a precise answer to the central question posed in the Introduction. The fundamental bottleneck in stochastic polynomial root-finding is neither the number of roots, nor the degree of the polynomial, nor the separation between roots per se, but a single scalar: the maximal saddle height H^* . This quantity governs the exponential rate of metastability through the identity $t_{\text{mix}} \approx e^{\{\beta H^*\}}$, which holds regardless of implementation details.

This insight generalizes well beyond polynomial root-finding. In any nonconvex optimization problem where the objective function is smooth and the global minima are isolated, the Eyring–Kramers framework identifies the minimax saddle height as the universal controlling parameter. Our contribution is to make this abstract principle quantitative and explicit in the polynomial setting, providing computable formulas

for both H^* and the pre-exponential constant $C(P)$.

8.2 Optimal Annealing Theory

A central practical implication of Theorem 5.1 is a rigorous constraint on the annealing schedule $\beta(t)$. To achieve global exploration across all n roots, the schedule must satisfy two competing requirements: the temperature must be high enough for the chain to cross barriers, yet low enough for the Gibbs measure to concentrate near global minima.

Theorem 8.1 (Necessary Condition for Global Coverage)

Let $\beta : [0, T] \rightarrow \mathbb{R}_{>0}$ be a non-decreasing annealing schedule. For the Langevin chain to achieve ε -accurate global coverage of all n roots with high probability, it is necessary that

$$T \geq C(P, \varepsilon) \cdot \exp(\beta(T) \cdot H^*),$$

where $C(P, \varepsilon)$ is an explicit constant. Equivalently, any schedule satisfying the differential condition

$$d\beta/dt \ll \lambda_1(\beta(t)) = C(P) \cdot \exp(-\beta(t) H^*)$$

guarantees asymptotically optimal global exploration. Schedules that heat faster than this rate provably trap the chain in local basins.

The condition $d\beta/dt \ll C(P) \cdot e^{-\beta H^*}$ is a quantitative refinement of the classical condition $\beta(t) \geq c \log t$ of Geman–Geman [3]. The improvement is that the logarithmic schedule is a sufficient condition for global convergence, but Theorem 8.1 provides the necessary condition, and identifies H^* as the single parameter that must be estimated to design an optimal schedule. For the quintic $P(z) = z^5 - z - 1$, this gives the explicit threshold $d\beta/dt \ll 2.637 \cdot e^{-2.356\beta}$.

8.3 Implications for High-Dimensional Optimization

The dimension-dependent scaling law of Appendix B (Theorem B.2 and the m -dimensional Eyring–Kramers formula) reveals a structural feature of high-dimensional nonconvex optimization. The additional polynomial prefactor β^{m-1} in $\lambda_1(\beta) \sim C_m(F) \cdot \beta^{m-1} \cdot e^{-\beta H_m^*}$ represents a dimensionality dividend: in higher dimensions, the saddle geometry provides additional directions for the chain to navigate barriers, modestly improving mixing relative to the pure exponential. However, this polynomial improvement is negligible against the exponential $e^{-\beta H_m^*}$ factor, confirming that energy barriers remain the dominant bottleneck regardless of dimension.

This has direct implications for understanding gradient-based optimizers in deep learning. The loss landscapes of neural networks exhibit a qualitatively similar structure: global minima are separated by saddle points, and the effective “energy barrier” between basins governs how long stochastic gradient descent must run to escape local optima. Our framework provides a rigorous template for quantifying these barriers in a setting where the energy is a polynomial (or polynomial-approximable) function of the parameters.

8.4 Parallelization and Its Limits

A natural practical question is whether massively parallel implementations—running many Langevin chains simultaneously on GPU hardware—can circumvent the exponential mixing time. The answer is nuanced and important to state precisely.

Remark 8.2 (What parallelization achieves and does not achieve). Running K independent Langevin chains in parallel reduces the coverage time by a factor of K : if each chain has mixing time t_{mix} , then $K = O(n)$ chains suffice to cover all n roots in time t_{mix} / n with high probability. However, parallelization does **not** reduce the exponential dependence on βH^* : the mixing time of each individual chain is still $\Theta(e^{\beta H^*})$. GPU acceleration shortens wall-clock time by running many such chains simultaneously, but the fundamental information-theoretic lower bound of Contribution (C3) applies to any stochastic algorithm, parallel or sequential, that uses Langevin-type transitions. Overcoming the $e^{\beta H^*}$ barrier requires a fundamentally different algorithmic approach—such as Parallel Tempering, which enables inter-chain communication across temperature levels and is not captured by Theorem 5.1.

8.5 Open Problems

The following problems remain open:

- (O1) **Sharp annealing schedule:** Determine the schedule $\beta^*(t)$ that is simultaneously necessary and sufficient for ε -accurate global coverage, and characterize its dependence on H^* and $C(P)$.
- (O2) **Degenerate case:** When P has multiple roots, E is no longer Morse. Extend the analysis using Bott–Morse theory for the degenerate critical submanifolds; the Eyring–Kramers formula must be replaced by a more general metastability result.
- (O3) **Multivariate extension:** Complete the programme of Appendix B: prove the sharp Morse index bounds for Type-S critical points, characterize H^*_m via resultants, and derive the full mixing time formula for m -dimensional polynomial systems. See Appendix B for the precise research agenda.

Appendix B. Towards a Multivariate Extension (Open Problem O3)

This appendix formulates the precise research programme for the multivariate extension (O3) and identifies the key obstacles. It constitutes a detailed agenda for a companion paper.

B.1 Setting and Energy Function

Let $F = (F_1, \dots, F_m) : \mathbb{C}^m \rightarrow \mathbb{C}^m$ be a polynomial map with $F_k \in \mathbb{C}[z_1, \dots, z_m]$ of degree d_k . Identifying $\mathbb{C}^m \cong \mathbb{R}^{2m}$ via $z_k = x_k + iy_k$, define the energy

$$E : \mathbb{R}^{2m} \rightarrow \mathbb{R}_{\geq 0}, \quad E(\zeta) = \|F(\zeta)\|^2 = \sum_{k=1}^m |F_k(\zeta)|^2.$$

The global minima of E are exactly the common zeros of the system $F_1 = \dots = F_m = 0$. By Bézout's theorem, there are at most $D = d_1 \dots d_m$ such zeros (counted with multiplicity in $\mathbb{C}^m$), all of which achieve $E = 0$.

B.2 Gradient and Critical Point Structure

A direct computation gives the gradient of E in real coordinates. Writing $J_F(\zeta)$ for the $m \times m$ complex Jacobian matrix with entries $(J_F)_{k\ell} = \partial F_k / \partial z_\ell$, and viewing this as a $2m \times 2m$ real matrix $J_F^{\mathbb{R}}$ via the standard real embedding, we have

$$\nabla E(\zeta) = 2 (J_F^{\mathbb{R}}(\zeta))^T F^{\mathbb{R}}(\zeta),$$

where $F^{\mathbb{R}}$ is the real $2m$ -vector form of F . Consequently:

Proposition B.1 (Multivariate Critical Point Locus)

The critical points of E are of two types:

- (i) Type-M (minima): points ζ^* with $F(\zeta^*) = 0$. These have Morse index 0 whenever $J_F(\zeta^*)$ is invertible (i.e., at simple zeros).
- (ii) Type-S (saddles): points with $F(\zeta) \neq 0$ and $(J_F^{\mathbb{R}})^T F^{\mathbb{R}} = 0$. The condition $(J_F^{\mathbb{R}})^T F^{\mathbb{R}} = 0$ defines an algebraic variety of complex dimension $m - 1$ (generically), and the Morse index of such points ranges over $\{1, 2, \dots, 2m - 1\}$.

B.3 Bézout Bound on Critical Points

The Type-M critical points number at most $D = d_1 \dots d_m$ by Bézout. The Type-S critical points are solutions of the system

$$F_k(\zeta) = 0 \quad \text{ext\{ is replaced by \} } (J_F^{\mathbb{R}})^T F^{\mathbb{R}} = 0, \quad \zeta \in \mathbb{C}^m,$$

which is a system of $2m$ polynomial equations in $2m$ real variables of degrees at most $(d_k + d_k - 1)$. By the real Bézout theorem, the number of isolated Type-S critical points is bounded by

$$N_S \leq \prod_{k=1}^m (2d_k - 1)^2.$$

For the scalar case $m = 1$, $d_1 = n$, this gives $N_S \leq (2n-1)^2$, whereas the exact count from Proposition 3.2 is $n - 1$. This gap between the Bézout bound and the exact count in the scalar case suggests that the multivariate setting admits a sharper analysis exploiting the structure of E as a sum of squared norms—a key open question.

B.4 Hessian Structure at Type-M Points

The Morse-theoretic analysis of Type-M points generalizes cleanly. At a simple zero ζ^* of F , expanding $F(\zeta) \approx J_F(\zeta^*)(\zeta - \zeta^*)$ gives

$$E(\zeta) \approx \|J_F(\zeta^*)\delta\zeta\|^2 = \delta\zeta^T (J_F^{\mathbb{R}})^T J_F^{\mathbb{R}} \delta\zeta,$$

where $\delta\zeta = \zeta - \zeta^*$ is the $2m$ -real displacement. The Hessian is therefore

$$H_E(\zeta^*) = 2(J_F^{\mathbb{R}}(\zeta^*))^T J_F^{\mathbb{R}}(\zeta^*),$$

which is positive semidefinite. It is positive definite (Morse index 0) if and only if $J_F(\zeta^*)$ has full rank m —exactly the condition that ζ^* is a simple zero. This directly extends Proposition 3.2 (Case i) to all dimensions.

Theorem B.2 (Multivariate Morse Index at Simple Zeros)

If ζ^* is a simple zero of F (i.e., $F(\zeta^*) = 0$ and $\det J_F(\zeta^*) \neq 0$), then ζ^* is a non-degenerate local minimum of E with Morse index 0 and

$$\det H_E(\zeta^*) = 2^{2m} |\det J_F(\zeta^*)|^4 > 0.$$

In particular, for $m = 1$ this recovers $\det H_E(z^*) = 4|P'(z^*)|^2$ from Proposition 3.2.

B.5 The Saddle Height and Eyring–Kramers in m Dimensions

The Eyring–Kramers formula generalizes to $\mathbb{R}^{2m}$ as follows (Bovier–den Hollander [7], Theorem 5.1). Under the analogous non-degeneracy conditions, the spectral gap of the Langevin diffusion on $\mathbb{R}^{2m}$ with energy $E(\zeta) = \|F(\zeta)\|^2$ satisfies

$$\lambda_1(\beta) \sim C_m(F) \cdot \beta^{m-1} \cdot \exp(-\beta H_m^*),$$

where $H_m^* = \min_{\gamma} \max_{t \in [0,1]} E(\gamma(t))$ is the minimax saddle height over all paths γ connecting distinct minimizers, and the pre-exponential factor $C_m(F)$ now involves the ratio of $(2m \times 2m)$ Hessian determinants. The critical new feature relative to $m = 1$ is the β^{m-1} polynomial prefactor, which originates from the $(2m-1)$ -dimensional saddle geometry.

The principal open challenge for the multivariate setting is to identify the minimax saddle path. In the univariate case, the saddle between any two roots z_j^*, z_k^* is uniquely determined by the root of P' in the convex hull of $\{z_j^*, z_k^*\}$ (Gauss–Lucas). In the multivariate case, the analogous characterization requires the theory of discriminants and resultants for polynomial systems, connecting our framework to numerical algebraic geometry [Mehta, 2011; Chen–Mehta, 2015].

B.6 Illustrative Example: $m = 2$, Quadratic System

To ground the above, consider the simplest non-trivial case: $m = 2$, $d_1 = d_2 = 2$, with the system

$$F_1(z_1, z_2) = z_1^2 + z_2^2 - 1, \quad F_2(z_1, z_2) = z_1 z_2.$$

The common zeros are $(\pm 1, 0)$ and $(0, \pm i)$, giving $D = d_1 d_2 = 4$ zeros as predicted by Bézout. The Jacobian is

$$J_F(z_1, z_2) = \begin{bmatrix} 2z_1 & 2z_2 \\ z_2 & z_1 \end{bmatrix}, \quad \det J_F = 2(z_1^2 - z_2^2).$$

The Type-S critical points satisfy $(J_F^{\mathbb{R}})^T F^{\mathbb{R}} = 0$ with $F \neq 0$. In this case, $\det J_F = 0$ gives $z_1 = \pm z_2$, so the Type-S locus is the pair of lines $\{z_1 = z_2\} \cup \{z_1 = -z_2\}$. The saddle height H_2^* is the maximum of $\|F\|^2$ over this locus, intersected with the minimax path between minimizers—a quantity

that can be computed explicitly and yields

$$H^*_2 = \|F(\pm 1/\sqrt{2}, \pm 1/\sqrt{2})\|^2 = |1/2 + 1/2 - 1|^2 + |1/2|^2 = 1/4.$$

Theorem B.2 and the $m = 2$ Eyring–Kramers formula then give $\lambda_1(\beta) \sim C_2(F) \cdot \beta \cdot e^{-\beta/4}$, with the linear β prefactor being the new feature absent in the $m = 1$ case.

Research Agenda for Companion Paper

The following specific results are conjectured and constitute the open programme (O3):

- (a) **Sharp Morse index bounds:** For a generic polynomial map F of degrees $d_1, \dots, d_m$, all Type-S critical points of E have Morse index in $\{1, \dots, 2m-1\}$, with exact index determined by the signature of a structured $2m \times 2m$ bordered Hessian.
- (b) **Minimax path via resultants:** The minimax saddle height H^*_m admits an algebraic characterization as the minimum value of the resultant $\text{Res}(F_1, \dots, F_m)$ restricted to the critical locus.
- (c) **Mixing time in m dimensions:** The mixing time of Langevin MCMC for the system F satisfies $t_{\text{mix}}(\epsilon) = O(\beta^{1-m} e^{\beta H^*_m} \cdot \beta^{-(m-1)} \cdot \epsilon^{-1})$, connecting spectral geometry to numerical algebraic geometry.

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Appendix A. Full Proof of Theorem 5.1

We verify that the hypotheses of the Eyring–Kramers theorem of Bovier et al. [5] are satisfied for $E(z) = |P(z)|^2$.

A.1 Verification of Hypotheses.

(H1) Smoothness: $E \in C^\infty(\mathbb{R}^2)$ since P is a polynomial. ✓

(H2) Sub-level set compactness: For any $c > 0$, $\{z : E(z) \leq c\}$ is compact since $E(z) \rightarrow \infty$ as $|z| \rightarrow \infty$. ✓

(H3) Non-degenerate critical points: Established in Proposition 3.2 under the genericity assumption. ✓

A.2 Eyring–Kramers Asymptotics.

Under (H1)–(H3), Theorem 1.1 of Bovier et al. [5] gives, for the communication height between two adjacent minimizers z_a^*, z_b^* separated by saddle $w_{\{ab\}}$,

$$\lambda_1(\beta) \sim |\lambda^-(w_{\{ab\}})| / 2\pi \cdot \sqrt{(\det H_E(z_a^*)) / (\det H_E(w_{\{ab\}}))} \cdot e^{-\beta[E(w_{\{ab\}}) - E(z_a^*)]}.$$

Since $E(z_a^*) = 0$ for any root of P , and the global bottleneck is the saddle w^* achieving H^* , the leading-order contribution to the spectral gap is as stated in Theorem 5.1. The sub-leading corrections are $O(\beta^{-1})$ and do not affect the exponential rate. ■